

AaharSetu – Limitations and Future Scope

1. Introduction

AaharSetu is a technology-driven solution designed to reduce food wastage by connecting restaurants, NGOs, and volunteers. While the system offers an efficient and scalable approach to food redistribution, certain limitations exist due to real-world constraints and development boundaries.

Understanding these limitations is essential for improving the system and planning future enhancements. This document outlines the current limitations of AaharSetu and explores potential future scope for expansion and improvement.

2. Limitations of AaharSetu

2.1 Limited Real-Time Reliability

The system relies heavily on real-time coordination between multiple users such as restaurants, NGOs, and volunteers.

Challenges:

- Delay in NGO response
- Volunteer unavailability
- Network issues

Impact:

Food may not be picked up on time, leading to spoilage.

2.2 Food Safety Constraints

Ensuring food safety is one of the biggest challenges in food redistribution systems.

Issues:

- No direct control over food preparation hygiene
- Difficulty in detecting spoilage automatically
- Dependence on manual verification

Impact:

Risk of distributing unsafe food.

2.3 Dependence on User Honesty

The system depends on accurate information provided by users.

Risks:

- Incorrect food details
- Fake donations
- Misleading information

Impact:

Reduces system reliability and trust.

2.4 Volunteer Availability Issues

Volunteers play a key role in delivery.

Problems:

- Limited number of volunteers
- Irregular availability
- Lack of commitment

Impact:

Delays in pickup and delivery.

2.5 Internet Dependency

The platform requires internet connectivity for real-time updates.

Issues:

- Poor network in some areas
- App usability affected

Impact:

System performance may degrade in low-network zones.

2.6 Scalability Challenges

As the number of users increases, system complexity grows.

Concerns:

- Managing large datasets
- Handling multiple requests simultaneously
- Maintaining performance

2.7 Limited Automation in MVP

The initial version (MVP) focuses on basic functionality.

Limitations:

- Manual volunteer assignment
- Basic matching system
- Limited analytics

2.8 Lack of Government Integration

Currently, the system operates independently.

Missing:

- Government support
- Policy integration
- Public distribution linkage

2.9 Data Privacy and Security Risks

Handling user data requires strong security.

Risks:

- Unauthorized access
- Data misuse

2.10 Logistics and Transportation Constraints

Food delivery depends on physical movement.

Issues:

- Traffic delays
- Distance challenges
- Lack of transport resources

3. Future Scope of AaharSetu

3.1 AI-Based Smart Matching System

Future versions can include intelligent algorithms to improve matching.

Features:

- Predict best NGO based on need
- Optimize volunteer assignment

3.2 Predictive Food Demand Analysis

Use AI to predict where food will be needed.

Benefits:

- Better planning
- Reduced waste

3.3 Route Optimization

Implement advanced routing algorithms.

Features:

- Shortest path delivery
- Multi-pickup routes

3.4 Advanced Food Safety System

Enhance food quality detection.

Future Enhancements:

- AI image-based freshness detection
- IoT temperature sensors

3.5 Blockchain-Based Transparency

Use blockchain for secure tracking.

Benefits:

- Prevent fraud
- Ensure transparency

3.6 Mobile Application Development

Develop Android/iOS apps.

Benefits:

- Better accessibility
- Real-time usage

3.7 Government and NGO Integration

Collaborate with authorities.

Impact:

- Wider reach
- Better implementation

3.8 Expansion to Multiple Cities

Scale the system across regions.

Goal:

- Nationwide implementation

3.9 Social Impact Analytics Dashboard

Provide detailed insights.

Features:

- Meals saved
- Waste reduced
- Impact reports

3.10 Gamification and Incentives

Encourage participation.

Ideas:

- Reward points
- Leaderboards

3.11 Offline Support System

Enable limited offline functionality.

Example:

- SMS-based requests

3.12 Corporate CSR Integration

Partner with companies.

Benefits:

- Funding
- Increased donations

4. Conclusion

AaharSetu provides a strong foundation for solving food wastage through technology. However, real-world challenges such as food safety, logistics, and user coordination present limitations that must be addressed.

With advancements in AI, IoT, and system scalability, AaharSetu has the potential to evolve into a comprehensive, intelligent, and nationwide food redistribution platform. Future improvements will enhance efficiency, reliability, and social impact, making it a sustainable solution for hunger management.